



Tecnológico Nacional de México

Instituto Tecnológico de Saltillo

Computo en la nube

5.8 Instalación entorno LAMP

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Paso 1 Conectarse a la VM

```
diego2004@VM-Ubuntu: ~  
Microsoft Windows [Versión 10.0.26200.8246]  
(c) Microsoft Corporation. Todos los derechos reservados.  
C:\Users\diego>ssh -i "C:\Users\diego\OneDrive\Escritorio\Octavo Semestre\Computo en la nube\VM-Ubuntu_key.pem" diego2004@104.211.28.177
```

```
diego2004@VM-Ubuntu:~$
```

Paso 2 Actualizar el sistema

```
diego2004@VM-Ubuntu:~$ sudo apt update  
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease  
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]  
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]  
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]  
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]  
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]  
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]  
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]  
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]  
Get:12 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]  
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1920 kB]  
Get:14 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [346 kB]  
Get:15 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]  
Get:16 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17.1 kB]  
Get:17 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1685 kB]  
Get:18 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [324 kB]  
Get:19 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]  
Get:20 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.5 kB]  
Get:21 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3054 kB]
```

```
Fetchd 41.8 MB in 5s (8324 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
28 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
diego2004@VM-Ubuntu:~$ sudo apt upgrade -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Calculating upgrade... Done  
The following upgrades have been deferred due to phasing:  
  iproute2  
The following packages will be upgraded:  
  gir1.2-packagekitglib-1.0 jq libcap2 libcap2-bin libjq1 libntfs-3g89t64 libpackagekit-glib2-18 libpam-cap  
  libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 ntfs-3g openssh-client openssh-server  
  openssh-sftp-server packagekit packagekit-tools python3.12 python3.12-minimal snapd ubuntu-pro-client  
  ubuntu-pro-client-l10n vim vim-common vim-runtime vim-tiny xxd  
27 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.  
24 standard LTS security updates
```

```

Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
/etc/needrestart/restart.d/systemd-manager
systemctl restart chrony.service multipathd.service polkit.service rsyslog.service systemd-journald.service systemd-net
workd.service systemd-resolved.service systemd-udev.service udisks2.service walinuxagent.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
diego2004 @ session #1: apt[1645], sshd[1147]
diego2004 @ user manager service: systemd[1152]
diego2004@VM-Ubuntu:~$

```

Paso 3 Instalar Apache

```

diego2004@VM-Ubuntu:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 1 not upgraded.
Need to get 2091 kB of archives.
After this operation, 8113 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9116 B]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.11 [1335 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.11 [163 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.11 [98.2 kB]

```

```

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

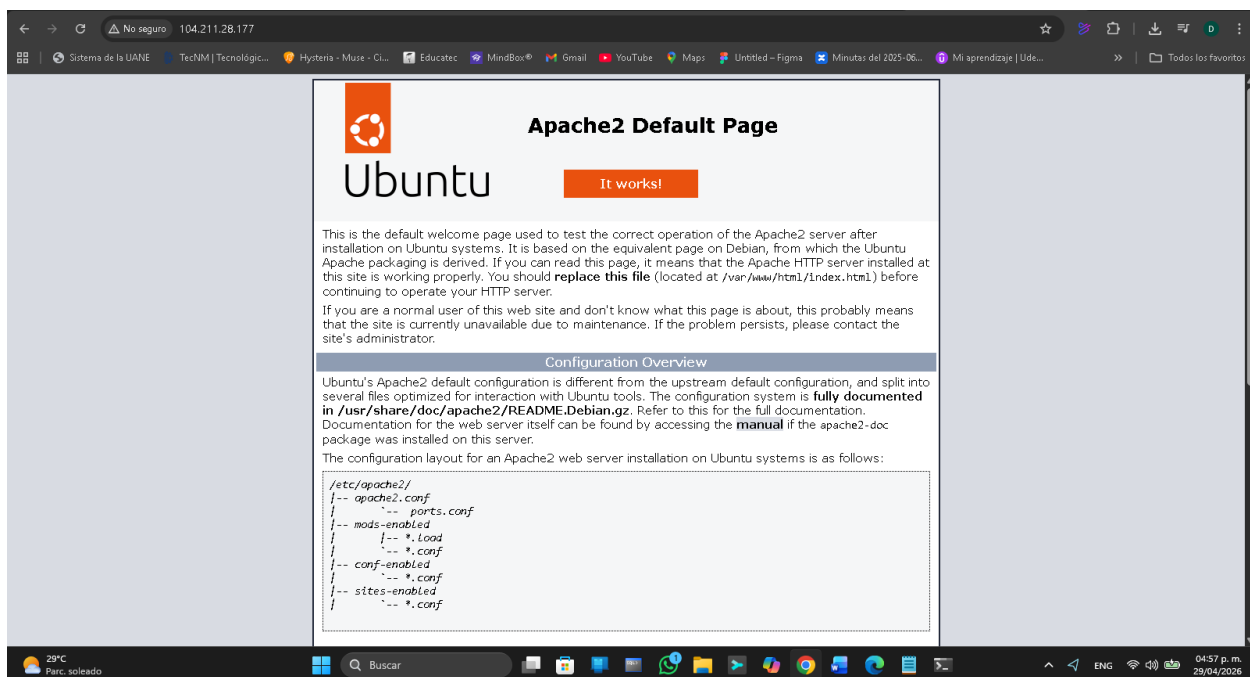
User sessions running outdated binaries:
diego2004 @ session #1: sshd[1147]
diego2004 @ user manager service: systemd[1152]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
diego2004@VM-Ubuntu:~$

```

```
diego2004@VM-Ubuntu:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-29 22:55:45 UTC; 54s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 7466 (apache2)
      Tasks: 55 (limit: 9504)
     Memory: 5.2M (peak: 5.3M)
        CPU: 23ms
    CGroup: /system.slice/apache2.service
            └─7466 /usr/sbin/apache2 -k start
              └─7469 /usr/sbin/apache2 -k start
                └─7470 /usr/sbin/apache2 -k start

Apr 29 22:55:45 VM-Ubuntu systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 29 22:55:45 VM-Ubuntu systemd[1]: Started apache2.service - The Apache HTTP Server.
```



Paso 4 Instalar PHP

```
diego2004@VM-Ubuntu:~$ sudo apt install php libapache2-mod-php php-mysql -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache2-mod-php8.3 php-common php8.3 php8.3-cli php8.3-common php8.3-mysql php8.3-opcache php8.3-readline
Suggested packages:
  php-pear
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.3 php php-common php-mysql php8.3 php8.3-cli php8.3-common php8.3-mysql
  php8.3-opcache php8.3-readline
0 upgraded, 11 newly installed, 0 to remove and 1 not upgraded.
Need to get 5051 kB of archives.
After this operation, 22.9 MB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 php-common all 2:93ubuntu2 [13.9 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-common amd64 8.3.6-0ubuntu0.24.04.8 [740 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-opcache amd64 8.3.6-0ubuntu0.24.04.8 [371 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-readline amd64 8.3.6-0ubuntu0.24.04.8 [13.5
```

Restarting services...

Service restarts being deferred:

```
/etc/needrestart/restart.d/dbus.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service
```

No containers need to be restarted.

User sessions running outdated binaries:

```
diego2004 @ session #1: sshd[1147]
diego2004 @ user manager service: systemd[1152]
```

No VM guests are running outdated hypervisor (qemu) binaries on this host.

diego2004@VM-Ubuntu:~\$

```
diego2004@VM-Ubuntu:~$ php -v
PHP 8.3.6 (cli) (built: Mar 20 2026 02:32:55) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.3.6, Copyright (c) Zend Technologies
    with Zend OPcache v8.3.6, Copyright (c), by Zend Technologies
diego2004@VM-Ubuntu:~$ |
```

Paso 5 Instalar MySQL

```
diego2004@VM-Ubuntu:~$ sudo apt install mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin
  libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtimedate-perl
  liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common
  mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl libregexp-ipv6-perl
  libwww-perl mailx tinycal
The following NEW packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin
  libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtimedate-perl
  liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common
  mysql-server mysql-server-8.0 mysql-server-core-8.0
0 upgraded, 28 newly installed, 0 to remove and 1 not upgraded.
Need to get 29.6 MB of archives.
After this operation, 243 MB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 mysql-common all 5.8+1.1.0build1 [6746 B]
```

Restarting services...

Service restarts being deferred:

```
/etc/needrestart/restart.d/dbus.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service
```

No containers need to be restarted.

User sessions running outdated binaries:

```
diego2004 @ session #1: sshd[1147]
diego2004 @ user manager service: systemd[1152]
```

No VM guests are running outdated hypervisor (qemu) binaries on this host.

diego2004@VM-Ubuntu:~\$

```
diego2004@VM-Ubuntu:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-29 23:00:22 UTC; 59s ago
     Process: 15849 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
    Main PID: 15858 (mysqld)
      Status: "Server is operational"
        Tasks: 38 (limit: 9504)
       Memory: 362.8M (peak: 380.6M)
          CPU: 685ms
       CGroup: /system.slice/mysql.service
               └─15858 /usr/sbin/mysqld

Apr 29 23:00:21 VM-Ubuntu systemd[1]: Starting mysql.service - MySQL Community Server...
Apr 29 23:00:22 VM-Ubuntu systemd[1]: Started mysql.service - MySQL Community Server.
diego2004@VM-Ubuntu:~$
```

```
diego2004@VM-Ubuntu:~$ sudo mysql_secure_installation
```

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

VALIDATE PASSWORD COMPONENT can be used to test passwords and improve security. It checks the strength of password and allows the users to set only those passwords which are secure enough. Would you like to setup VALIDATE PASSWORD component?

Press y|Y for Yes, any other key for No: y

There are three levels of password validation policy:

LOW Length >= 8

MEDIUM Length >= 8, numeric, mixed case, and special characters

STRONG Length >= 8, numeric, mixed case, special characters and dictionary file

Please enter 0 = LOW, 1 = MEDIUM and 2 = STRONG: 2

Skipping password set for root as authentication with auth_socket is used by default.

If you would like to use password authentication instead, this can be done with the "ALTER_USER" command.

See <https://dev.mysql.com/doc/refman/8.0/en/alter-user.html#alter-user-password-management> for more information.

By default, a MySQL installation has an anonymous user, allowing anyone to log into MySQL without having to have a user account created for them. This is intended only for testing, and to make the installation go a bit smoother.

Remove test database and access to it? (Press y|Y for Yes, any other key for No) : y

- Dropping test database...

Success.

- Removing privileges on test database...

Success.

Reloading the privilege tables will ensure that all changes made so far will take effect immediately.

Reload privilege tables now? (Press y|Y for Yes, any other key for No) : y

Success.

All done!

```
diego2004@VM-Ubuntu:~$
```

Paso 6 Probar servidor web

```
diego2004@VM-Ubuntu:~$ sudo nano /var/www/html/info.php
diego2004@VM-Ubuntu:~$
```

```
GNU nano 7.2
<?php
phpinfo();
?>
```

